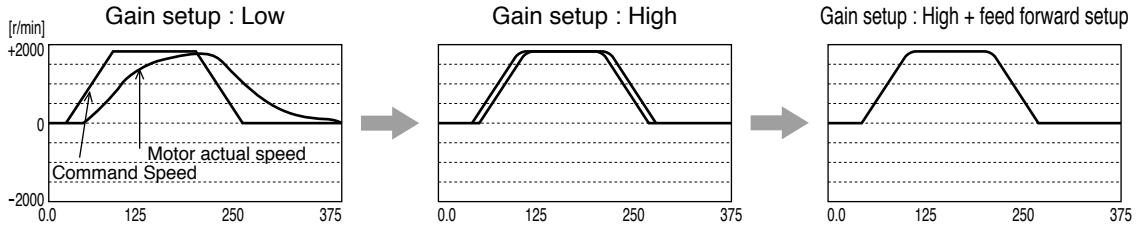
1. Gain Adjustment

Outline

Purpose

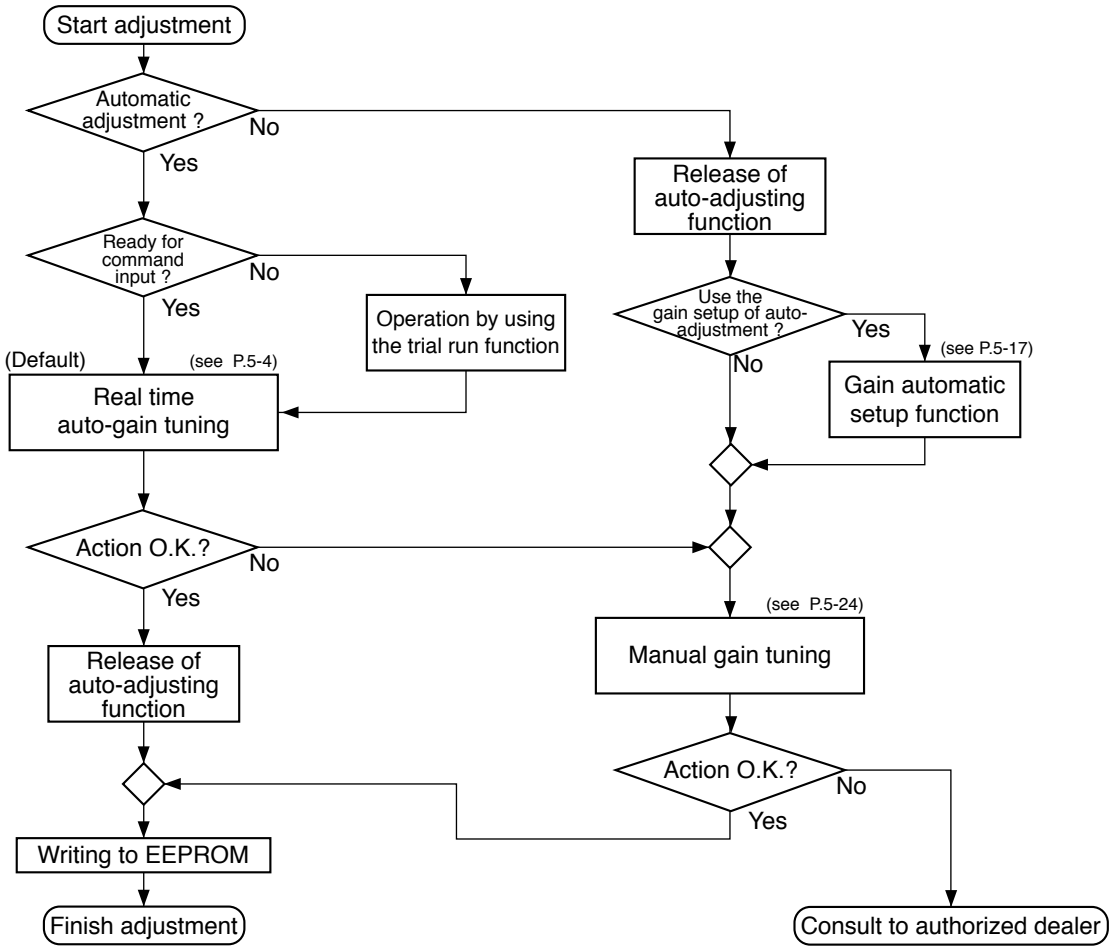
It is required for the servo driver to run the motor in least time delay and as faithful as possible against the commands from the host controller. You can make a gain adjustment so that you can run the motor as closely as possible to the commands and obtain the optimum performance of the machine.

<e.g. : Ball screw>



Gain setup : Low		Gain setup : High		Gain setup : High + feed forward setup	
Position loop gain	: 20	Position loop gain	: 100	Position loop gain	: 100
Velocity loop gain	: 100	Velocity loop gain	: 50	Velocity loop gain	: 50
Time constant of V-loop integration	: 50	Time constant of V-loop integration	: 50	Time constant of V-loop integration	: 50
Velocity loop feed forward	: 0	Velocity loop feed forward	: 0	Velocity loop feed forward	: 500
Inertia ratio	: 100	Inertia ratio	: 100	Inertia ratio	: 100

Procedures



Note For safety operation, first adjust the gain by referring to P.6-15 Setup protective function before gain adjustment.

Type

Function			Explanation	Pages to refer
Automatic adjustment	Real-time auto-gain tuning		Estimates the load inertia of the machine in real time, and automatically sets up the optimum gain corresponding to this result.	P.5-4
	Adaptive filter		Reduces the resonance vibration point by automatically setting up the notch filter coefficient which removes the resonance component from the torque command while estimating the resonance frequency from the vibrating component which appears in the motor speed in actual operating condition.	P.5-10
Manual adjustment	Manual gain tuning (basic)		Execute the manual adjustment or fine-tuning when real-time auto-gain tuning cannot be activated due to the limitation of operation or load condition, or when you want to obtain an optimum response and stability under these conditions.	P.5-13
		Basic procedure	Adjustment of position control mode	P.5-14
			Adjustment of velocity control mode	P.5-15
			Adjustment of torque control mode	P.5-15
			Adjustment of full-closed control mode	P.5-16
		Gain switching function	You can expect to reduce vibration at stopping and settling time and to improve command compliance by switching the gains by internal data or external signals.	P.5-17
		Suppression of machine resonance	When the machine stiffness is low, vibration or noise may be generated due to the distorted axis, hence you cannot set the higher gain. You can suppress the resonance with two kinds of filter.	P.5-20
	Manual gain tuning (application)		You can obtain the higher performance while you are not satisfied with the performance obtained with the basic adjustment, using the following application functions.	
		Damping control	Function which reduces vibration by removing the vibration frequency component while the front end of the machine vibrates.	P.5-24
		Feed forward function	Velocity feed forward function improves responsiveness during position control and full closed control. Torque feed forward improves the response of velocity control system.	P.5-26
		Instantaneous speed observer	Function which obtains both high response and reduction of vibration at stopping by estimating the motor speed with the load model, and hence improves the accuracy of speed detection.	P.5-28
		Disturbance observer	Function which uses estimated disturbance torque to reduce effects of the disturbance torque and to reduce vibration.	P.5-30
		3rd gain switching function	By using this function in addition to the normal gain switching function, the gain can be changed at the moment of stop to further shorten the positioning time.	P.5-32
		Friction torque compensation	Offset load compensation and dynamic friction compensation are used to reduce effects of mechanical friction.	P.5-34
		Inertia ratio switching function	This function can be used when selectable 2 inertia ratios are provided.	P.5-36
		Hybrid vibration damping function	This function, when used in full closed control mode, prevents vibration resulting from torsion on motor and load.	P.5-38

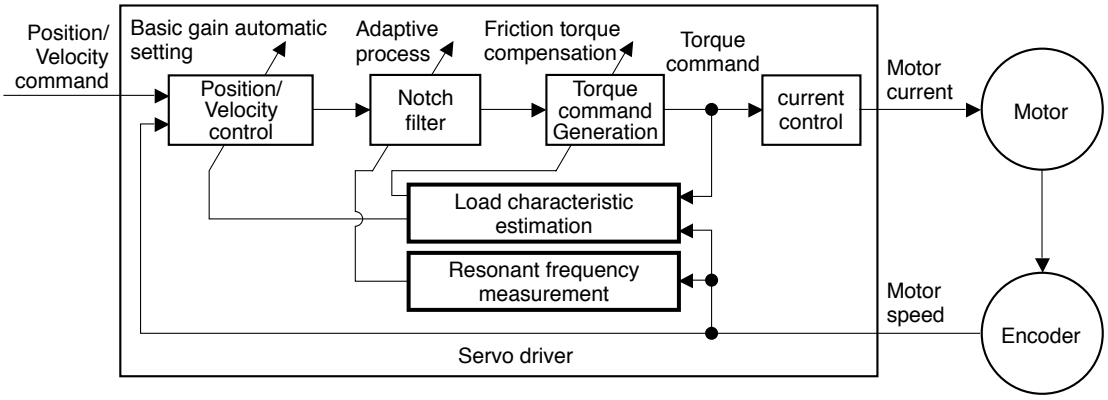
Remarks • Pay extra attention to safety, when oscillation (abnormal noise and vibration) occurs, shut off the main power, or turn to Servo-OFF.

2. Real-Time Auto-Gain Tuning

Basic

Outline

The system estimates the load characteristics in real time, and automatically performs basic gain setting and friction compensation by referring to stiffness parameter.



Applicable Range

Real time auto-gain tuning is applicable to all control modes.

	Real-time auto-tuning condition
Control Mode	Specific real-time auto-tuning mode is selected according to the currently active control mode. For details, refer to the description of Pr0.02 Real-time auto-tuning setup.
Others	• Should be in servo-on condition • Input signals such as the deviation counter clear and command input inhibit, and parameters except for controls such as torque limit setup, are correctly set, assuring that the motor can run smoothly.

Caution

Real-time auto-gain tuning may not be executed properly under the conditions described in the table below. Under these conditions, change the load condition or operation pattern, or start manual gain tuning (see P.5-24).

	Conditions which obstruct real-time auto-gain tuning action
Load inertia	• The load is too small or large compared to the rotor inertia. (less than 3 times or more than 20 times). • The load inertia changes too quickly.
Load	• The machine stiffness is extremely low. • Nonlinear characteristics such as backlash exist.
Action pattern	• The motor is running continuously at low speed of (100 [r/min] or lower. • Acceleration/deceleration is slow (2000 [r/min] per 1[s] or low). • Acceleration/deceleration torque is smaller than unbalanced weighted/viscous friction torque. • When the speed condition of 100 [r/min] or more and acceleration/deceleration condition of 2000 [r/min] per 1 [s] are not maintained for 80 [ms].

How to Operate

1) Bring the motor to stall (Servo-OFF).
2) Set up Pr0.02 (Setup of real-time auto-gain tuning mode) to 1-7.
   Default is set to 1.

Setup value	Real-time auto-gain tuning
0	Invalid
1	Standard
2	Positioning *1
3	Vertical axis *2
4	Friction compensation *3
5	Load characteristic measurement
6	Customize *4

*1 Velocity and torque controls are the same as in the standard mode.
*2 Torque control is the same as in the standard mode.
*3 Velocity control is the same as in the vertical axis mode. Torque control is the same as in the standard mode.
*4 Certain function(s) is not available in a specific control mode. Refer to description in Pr6.32.

   Control parameter is automatically set according to Pr0.03 Real-time auto-tuning stiffness setup. For details, see P.5-6 and 5-7.
3) Turn on servo, and start the machine.

   Estimation of load characteristics starts.
4) When the load characteristics are determined, Pr0.04 Inertia ratio is updated.
   In a specific mode, the following parameters are changed:
   Pr6.07 Torque command additional value
   Pr6.08 Positive direction torque compensation value
   Pr6.09 Negative direction torque compensation value
5) When value of Pr0.03 Real-time auto-tuning stiffness setup is increased, the motor responsiveness will be improved.
   Determine the most appropriate stiffness in relation to the positioning setup time and vibration condition.
6) To save the result to memory, write the data to EEPROM.

Caution If power is turned off within 30 minutes after the end of tuning process, the result of the real-time auto-tuning is not saved. If the result is not saved, manually write parameters to EEPROM and then turn off power.

Note
- While the auto-tuning is valid, parameters that are to be automatically adjusted cannot be changed.

Related page
- P.2-71 “EEPROM Writing”
- P.4-5, 4-6, 4-51, 4-52, 4-55, 4-56 “Details of parameter”

Parameters set/changed by real-time auto-gain tuning

• Parameters which are updated

The real-time auto-tuning function updates the following parameters according to Pr0.02 Real-time auto-tuning setup and Pr6.32 Real-time auto-tuning custom setup and by using the load characteristic estimate values.

Class	No.	Title	Function
0	04	Inertia ratio	Updates this parameter when the real-time auto-tuning inertia ratio update is enabled.
6	07	Torque command additional value	Update this parameter when the vertical axis mode for real time auto-tuning is valid.
6	08	Positive direction torque compensation value	Update this parameter when the friction compensation mode for real time auto-tuning is valid.
6	09	Negative direction torque compensation value	Update this parameter when the friction compensation mode for real time auto-tuning is valid.

• Parameters which are updated to setup value corresponding to stiffness setup

The real-time auto-tuning function updates the following basic gain setup parameters according to Pr0.03 Real-time auto-tuning stiffness setup.

Class	No.	Title	Function
1	00	1st gain of position loop	When stiffness setup is valid, updates the parameter based on the setup value. Refer to P.5-9 Basic gain parameter setup table.
1	01	1st gain of velocity loop	
1	02	1st time constant of velocity loop integration	
1	04	1st time constant of torque filter	
1	05	2nd gain of position loop	
1	06	2nd gain of velocity loop	
1	07	2nd time constant of velocity loop integration	
1	09	2nd time constant of torque filter	

• Parameters which are set to fixed value

Real-time auto-tuning function sets the following parameters to the fixed value.

Class	No.	Title	Setup value when fixed parameter setup is valid.
1	03	1st filter of speed detection	0
1	08	2nd filter of speed detection	
1	10	Velocity feed forward gain	300 (30%)
1	11	Velocity feed forward filter	50 (0.5ms)
1	12	Torque feed forward gain	0
1	13	Torque feed forward filter	

Related page • P.4-6 “Pr0.04” • P.4-12... “Pr1.00...” • P.4-52 “Pr6.07...”

• Parameters which are set in response to gain switching setup

The real-time auto-tuning function sets the following parameters as the gain is switched.

Class	No.	Title	Function
1	14	2nd gain setup	Sets to 1 if the current setting is not maintained.
1	15	Mode of position control switching	Sets to 10 to enable the gain switching. Sets to 0 to disable the gain switching.
1	16	Delay time of position control switching	Sets to 50 if the current setting is not maintained.
1	17	Level of position control switching	
1	18	Hysteresis at position control switching	Sets to 33 if the current setting is not maintained.
1	19	Position gain switching time	
1	20	Mode of velocity control switching	Sets to 0 if the current setting is not maintained.
1	21	Delay time of velocity control switching	
1	22	Level of velocity control switching	
1	23	Hysteresis at velocity control switching	
1	24	Mode of torque control switching	
1	25	Delay time of torque control switching	
1	26	Level of torque control switching	
1	27	Hysteresis at torque control switching	

• Parameters which are always set to invalid.

The following settings are always set to invalid when Pr0.02 Real-time auto-tuning setup is not 0.

Class	No.	Title	Function
6	10	Function expansion setup	Instantaneous speed observer function enable bit (bit 0), disturbance observer function enable bit (bit 1) and inertia ratio switching function enable bit (bit 3) are internally disabled.
6	13	2nd Inertia ratio	Parameter setup can be changed, but disturbance observer is disabled.
6	23	Disturbance torque compensating gain	
6	24	Disturbance observer filter	

Related page • P.4-14... “Pr1.14...” • P.4-52... “Pr6.10...”

Caution

(1) Immediately after the first servo-on upon start up; or after increasing Pr0.03 Real-time auto-tuning stiffness setup, abnormal sound or oscillation may be generated until the load characteristics estimation is stabilized. If such abnormality lasts or repeats for 3 or more reciprocating operations, take the following countermeasures.
   1) Lower the setup of Pr0.03 (Selection of machine stiffness at real-time auto-gain tuning).
   2) Set Pr0.02 Real-time auto-tuning setup to 0 to disable the real-time auto-tuning.
   3) Set Pr0.04 Inertial ratio to the calculational value of the equipment and set Pr6.07 Torque command addition value, Pr6.08 Positive direction compensation value and Pr6.09 Negative direction compensation value to 0.

(2) When abnormal noise and oscillation occur, Pr0.04 (Inertia ratio) or Pr6.07 (Torque command additional value), Pr6.08(Positive direction torque compensation value), Pr6.09(Negative direction torque compensation value) might have changed to extreme values. Take the same measures as the above in these cases.

(3) Among the results of real-time auto-gain tuning, Pr0.04 (Inertia ratio) and Pr6.07 (Torque command additional value), Pr6.08(Positive direction torque compensation value), Pr6.09(Negative direction torque compensation value) will be written to EEPROM every 30 minutes. When you turn on the power again, the auto-gain tuning will be executed using the latest data as initial values.

Invalidation of Real-Time Auto-Gain Tuning

You can stop the automatic calculation of Pr0.04 (Inertial ratio) and invalidate the real-time auto-gain tuning by setting up Pr0.02 (Real-time auto-gain tuning setup) to 0.
Note that the calculation result of Pr0.04 (Inertia ratio) will be held, and if this parameter becomes abnormal value, use the normal mode auto-gain tuning or set up proper value manually obtained from formula or calculation.

Caution If power is turned off within 30 minutes after the end of tuning process, the result of the real-time auto-tuning is not saved. If the result is not saved, manually write parameters to EEPROM and then turn off power.

Basic gain parameter setup table

Stiffness	1st gain				2nd gain				A4 Series Stiffness setup (reference) *1
	Pr1.00	Pr1.01	Pr1.02	Pr1.04 *2	Pr1.05	Pr1.06	Pr1.07	Pr1.09 *2	
	Gain of position loop [0.1/s]	Gain of velocity loop [0.1Hz]	Time constant of velocity loop integration [0.1ms]	Time constant of torque filter [0.01ms]	Gain of position loop [0.1/s]	Gain of velocity loop [0.1Hz]	Time constant of velocity loop integration [0.1ms]	Time constant of torque filter [0.01ms]	
0	20	15	3700	1500	25	15	10000	1500	—
1	25	20	2800	1100	30	20	10000	1100	—
2	30	25	2200	900	40	25	10000	900	—
3	40	30	1900	800	45	30	10000	800	—
4	45	35	1600	600	55	35	10000	600	—
5	55	45	1200	500	70	45	10000	500	—
6	75	60	900	400	95	60	10000	400	—
7	95	75	700	300	120	75	10000	300	—
8	115	90	600	300	140	90	10000	300	0
9	140	110	500	200	175	110	10000	200	—
10	175	140	400	200	220	140	10000	200	—
11	320	180	310	126	380	180	10000	126	1
12	390	220	250	103	460	220	10000	103	2
13	480	270	210	84	570	270	10000	84	3
14	630	350	160	65	730	350	10000	65	4
15	720	400	140	57	840	400	10000	57	5
16	900	500	120	45	1050	500	10000	45	6
17	1080	600	110	38	1260	600	10000	38	7
18	1350	750	90	30	1570	750	10000	30	8
19	1620	900	80	25	1880	900	10000	25	9
20	2060	1150	70	20	2410	1150	10000	20	10
21	2510	1400	60	16	2930	1400	10000	16	11
22	3050	1700	50	13	3560	1700	10000	13	12
23	3770	2100	40	11	4400	2100	10000	11	13
24	4490	2500	40	9	5240	2500	10000	9	14
25	5000	2800	35	8	5900	2800	10000	8	—
26	5600	3100	30	7	6500	3100	10000	7	15
27	6100	3400	30	7	7100	3400	10000	7	—
28	6600	3700	25	6	7700	3700	10000	6	—
29	7200	4000	25	6	8400	4000	10000	6	—
30	8100	4500	20	5	9400	4500	10000	5	—
31	9000	5000	20	5	10500	5000	10000	5	—

*1 Stiffness setting of A4 series refers to the setup value (0-15) of A4 series parameter Pr22 Real-time auto-tuning machine stiffness selection.

*2 When 17-bit absolute encoder, limited by the minimum value 10.

Note

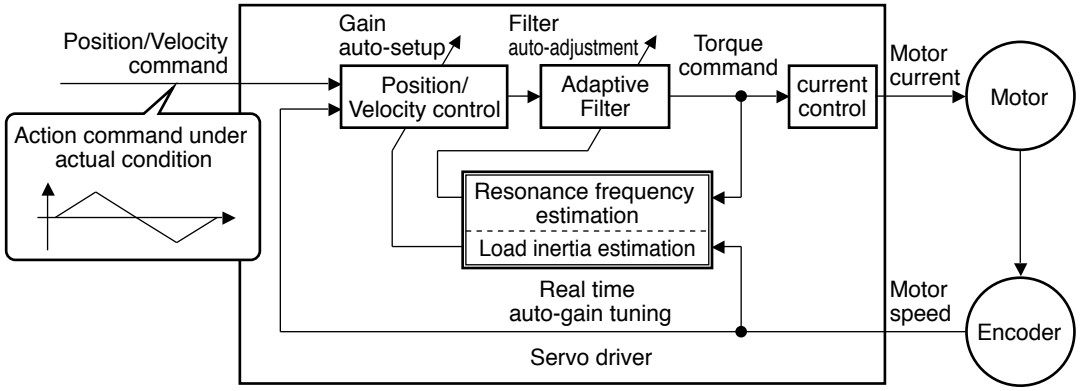
- For details of parameters, refer to P.4-12 “Details of parameter”.
- Download the A4 series manual from the web site shown below.
  http://industrial.panasonic.com/jp/i/25000/motor_fa/motor_fa.html

3. Adaptive filter

Adaptive filter

Outline

Estimates the resonance frequency out of vibration component presented in the motor speed in motion, then removes the resonance component from the torque command by setting up the notch filter coefficient automatically, hence reduces the resonance vibration.



Applicable Range

This function works under the following condition.

	Conditions under which the Adaptive filter is activated
Control Mode	• Applies to other control modes than torque control.
Others	• Should be servo-on status. • Elements other than control parameters, such as deviation counter clear command inhibit and torque limit are appropriately set, enabling the motor to run normally.

Caution

In the following condition, normal operation may not be expected - manually set the notch filter to prevent resonance.

	Conditions which obstruct adaptive filter action
Resonance point	• Resonance frequency is lower than 300[Hz]. • Resonance peak is low, or control gain is low where the motor speed is not affected by this. • Multiple resonance points exist.
Load	• Motor speed variation with high harmonic component is generated due to non-linear factors such as backlash.
Command pattern	• Acceleration/deceleration is rapid such as 30000[r/min] per 1[s].

How to Operate

Enter the action command with Pr2.00 Adaptive filter mode set to a value other than 0.

If the resonance point affects the motor speed, parameters of 3rd notch filter and/or 4th notch filters are automatically set according to the number of adaptive filters.

et the operation of the adaptive filter to the following parameter.

Class	No.	Title	Setup value	Function
2	00	Adaptive filter mode setup	0	[Adaptive filter: invalid] The adaptive filter is disabled. Parameters related to the 3rd and 4th notch filter hold the current value.
			1	[Adaptive filter: 1 filter is valid] One adaptive filter is enabled. Parameters related to the 3rd notch filter will be updated based on adaptive performance.
			2	[Adaptive filter: 2 filters are valid] Two adaptive filters are enabled. Parameters related to the 3rd and 4th notch filters will be updated based on adaptive performance.
			3	[Resonance frequency measurement mode] Measure the resonance frequency. Result of measurement can be checked with PANATERM. Parameters related to the 3rd and 4th notch filter hold the current value.
			4	[Clear result of adaptation] Parameters related to the 3rd and 4th notch filter are disabled and results of adaptive operation are cleared.

At the same time, the following parameters are automatically set.

Class	No.	Title	Function
2	07	3rd notch frequency	In no resonance point is found, the frequency is set to 5000.
2	08	3rd notch width selection	Automatically set when the adaptive filter is active.
2	09	3rd notch depth selection	
2	10	4th notch frequency	Notch frequency is automatically set to the 2nd resonance frequency estimated by the adaptive filter. In no resonance point is found, the frequency is set to 5000.
2	11	4th notch width selection	Automatically set when 2 adaptive filters are active.
2	12	4th notch depth selection	

Related page • P.4-19... “Details of parameter”

Caution

(1) Immediately after the first servo-on at start up; or after increasing stiffness setting with the real-time auto-tuning enabled, abnormal sound or oscillation may be generated until the adaptive filter stabilizes. If such abnormality lasts or repeats for 3 or more reciprocating operations, take the following countermeasures.
   1) Write the parameters which have given the normal operation into EEPROM.
   2) Lower the setup of Pr0.03 (Selection of machine stiffness at real-time auto-gain tuning).
   3) Invalidate the adaptive filter by setting up Pr2.00 (Setup of adaptive filter mode) to 0. (Reset of inertia calculation and adaptive action)
   4) Set up the notch filter manually.

(2) Abnormal sound or oscillation may excessively change the setup value of 3rd and 4th notch filters. If such change occurs, disable the adaptive filter as described in step 3) above, change setup value of Pr2.07 3rd notch frequency and Pr2.10 4th notch frequency to 5000 (disable), and then enable the adaptive filter again.

(3) The 3rd filters (Pr2.07-Pr2.09) and 4th notch filters (Pr2.10-Pr2.12) are written to EEPROM every 30 minutes. Upon power up, these data are used as default values during adaptive process.

Related page • P.4-5... “Details of parameter”

4. Manual Gain Tuning (Basic)

Outline

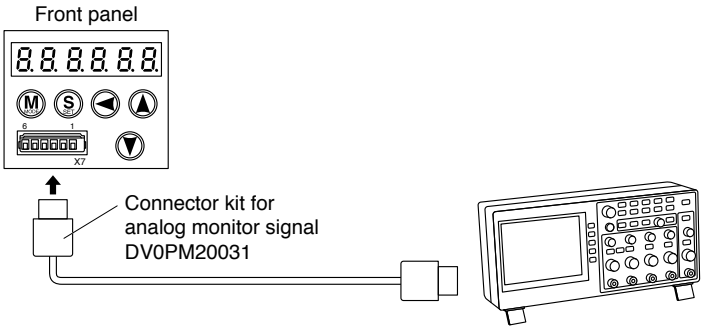
As explained previously, MINAS-A4 series features the automatic gain tuning function, however, there might be some cases where this automatic gain tuning cannot be adjusted properly depending on the limitation on load conditions. Or you might need to readjust the tuning to obtain the optimum response or stability corresponding to each load.
Here we explain this manual gain tuning method by each control mode and function.

Before Making a Manual Adjustment

By monitoring waveforms using the waveform graphic function of the setup support software PANATERM installed on the PC or by measuring the analog voltage waveform with the help of the monitor function, accurate adjustment can be positively, quickly and easily done when compared with that performed on the front panel.

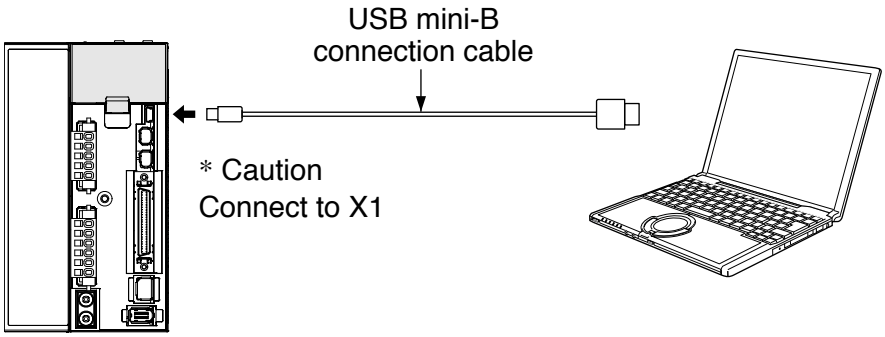
1. Analog monitor output

You can measure the actual motor speed, commanded speed, torque and deviation pulses by analog voltage level by using an oscilloscope. Set up the types of the signals or the output voltage level with Pr4.16 (Selection of speed monitor) and Pr4.21 (Selection of torque monitor).



2. Waveform graphic function of the PANATERM

You can display the command to the motor, motor movement (speed, torque command and deviation pulses) as a waveform graphic on PC display. Refer to P.7-26, "Outline of the Setup Support Software, PANATERM" of Supplement.



Caution Please download the Setup support software “PANATERM”from our web site anduse after install to the PC.

Related page • P.4-35, 4-36... “Details of parameter” • P.7-26 “Outline of Setup support software “PANATERM”

4. Manual Gain Tuning (Basic)

Adjustment in Position Control Mode

Position control of MINAS-A5 series is described in Block diagram of P.3-40.
Make adjustment in position control per the following procedures.

(1) Set up the following parameters to the values of the table below.

Parameter No. (Pr□□)	Title of parameter	Standard value
1.00	1st gain of position loop	270
1.01	1st gain of velocity loop	150
1.02	1st time constant of velocity loop integration	370
1.03	1st filter of velocity detection	0
1.04	1st time constant of torque filter time	152
1.10	Velocity feed forward	0
1.11	Time constant of feed forward filter	0
1.05	2nd gain of position loop	270
1.06	2nd gain of velocity loop	150
1.07	2nd time constant of velocity loop integration	370
1.08	2nd filter of speed detection	0
1.09	2nd time constant of torque filter	152
2.01	Selection of 1st notch frequency	5000
2.02	Selection of 1st notch width	2

Parameter No. (Pr□□)	Title of parameter	Standard value
0.04	Inertia ratio	100
0.02	Setup of real time auto-gain tuning mode	0
2.00	Adaptive filter setup mode	0
2.14	1st damping frequency	0
2.15	Setup of 1st damping filter	0
2.16	2nd damping frequency	0
2.17	Setup of 2nd damping filter	0
1.14	2nd gain setup	0
1.15	Mode of position control switching	0
1.16	Delay time of position control switching delay	0
1.17	Level of position control switching	0
1.18	Hysteresis at position control switching	0
1.19	Position gain switching time	0
2.22	Setup of smoothing filter	1
2.23	Setup of FIR filter	0

(2) Enter the inertia ratio of Pr20. Measure the ratio or setup the calculated value.
(3) Make adjustment using the standard values below.

Order	Parameter No. (Pr□□)	Title	Standard value	How to adjust
1	Pr11	1st gain of velocity loop	300	Increase the value within the range where no abnormal noise and no vibration occur. If they occur, lower the value.
2	Pr14	1st time constant of torque filter	50	When vibration occurs by changing Pr11, change this value. Setup so as to make Pr11 x Pr14 becomes smaller than 10000. If you want to suppress vibration at stopping, setup larger value to Pr14 and smaller value to Pr11. If you experience too large vibration right before stopping, lower than value of Pr14.
3	Pr10	1st gain of position loop	500	Adjust this observing the positioning time. Larger the setup, faster the positioning time you can obtain, but too large setup may cause oscillation.
4	Pr12	1st time constant of velocity loop integration	250	Setup this value within the range where no problem occurs. If you setup smaller value, you can obtain a shorter positioning time, but too small value may cause oscillation. If you setup too large value, deviation pulses do not converge and will be remained. Increase the value within the range where no abnormal noise occurs.
5	Pr15	Velocity feed forward	300	Too large setup may result in overshoot or chattering of position complete signal, hence does not shorten the settling time. If the command pulse is not even,you can improve by setting up Pr16 (Feed forward filter) to larger value.

Related page • P.4-5... “Details of parameter” • P.3-14... “Control Block Diagram”

4. Manual Gain Tuning (Basic)

Adjustment in Velocity Control Mode

Velocity control of MINAS-A5 series is described in Block Diagram of P.3-15 of Velocity Control Mode.
Adjustment in velocity control is almost same as that in position control described in "Adjustment in Position Control Mode", and make adjustments of parameters per the procedures except the gain setup of position loop and the setup of velocity feed forward.

5 Adjustment

4. Manual Gain Tuning (Basic)

Adjustment in Torque Control Mode

Torque control of MINAS-A5 series is described in P.3-16, "Block Diagram" of Torque Control Mode.
This torque control is based on velocity control while making Pr3.21 [Speed limit value 1], Pr3.22 [Speed limit value 2] or SPR input as a speed limit. Here we explain the setup of speed limiting value.

• Setup of speed limiting value

The torque command selection (Pr3.17) specifies the setup method.

Pr3.17 = 0	**Set up by using speed limit value 1 (Pr3.21)**
Pr3.17 = 1	**Set up by using analog input (SPL)**
Pr3.17 = 2	**For positive direction, set up by using the speed limit value 1 (Pr3.21)** **For negative direction, set up by using the speed limit value 2 (Pr3.22)**

- When the motor speed approaches to the speed limiting value, torque control following the analog torque command shifts to velocity control based on the speed limiting value.
- In order to stabilize the movement under the speed limiting, you are required to set up the parameters according to the above-mentioned "Adjustment in Velocity Control Mode".
- When the speed limiting value is too low or the velocity loop gain is too low, or when the time constant of the velocity loop integration is 10000 (invalid), the input to the torque limiting portion of the above fig. becomes small and the output torque may not be generated as the analog torque command.

Related page • P.3-43 “Inputs and outputs on connector X4”

4. Manual Gain Tuning (Basic)

Adjustment in Full-Closed Control Mode

Full-closed control of MINAS-A5 series is described in Block diagram of P.3-17 of Full-Closed Control.
Adjustment in full-closed control is almost same as that in position control described in P.3-12 “Adjustment in Position Control Mode”, and make adjustments of parameters per the procedures except cautions of P.5-14, “Outline of Full-Closed Control” (difference of command unit and difference of electronic gear).
Here we explain the setup of feedback scale ratio and hybrid deviation excess of full-closed control.

1) Setup of external scale ratio

Setup the external scale ratio using the numerator of external scale division (Pr3.24) and denominator of external scale division (Pr3.25).

- Check the encoder pulse counts per one motor revolution and the external scale pulse counts per one motor revolution, then set up the numerator of external scale division (Pr3.24), and denominator of external scale division (Pr3.25) so that the following formula can be established.

$$\frac{\text{Pr3.24}}{\text{Pr3.25}} = \frac{\text{Number of encoder pulses per motor rotation}}{\text{Number of external scale pulses per motor rotation}}$$

- If this ratio is incorrect, a gap between the position calculated from the encoder pulse counts and that of calculated from the external scale pulse counts will be enlarged and hybrid deviation excess (Err25.0) will be triggered when the work or load travels a long distance.
- When you set up Pr3.24 to 0, the encoder pulse counts will be automatically set up.

2) Setup of hybrid deviation excess

Set up the minimum value of hybrid deviation excess (Pr3.28) within the range where the gap between the motor (encoder) position and the load (feedback scale) position will be considered to be an excess.

- Note that the hybrid deviation excess (Err25.0) may be generated under other conditions than the above 1), such as reversed connection of the external scale or loose connection of the motor and the load.

Caution

(1) Enter the command pulses based on the feedback scale reference.
(2) The feedback scales to used for full-closed control are as follows.
When A- and B-phase parallel, or serial scale,
- ABS ST770A, ST770AL, AT573A series by Mitsutoyo Corp.
- SR77, SR87, SL700, SL710 by Sony Manufacturing Systems Corp.

(3) To prevent the runaway and damage of the machine due to the setup of the feedback scale, setup the hybrid deviation excess (Pr3.28) to the appropriate value, in the unit of feedback scale resolution.

(4) **We recommend the external scale as 1/40 ≤ external scale ratio ≤ 160.**

Even within this range, if you setup the external scale ratio to smaller value than 50/position loop gain (Pr1.00 and 1.05), you may not be able to control by one pulse unit. If you set up too large external scale ratio, you may expect larger noise in movement.

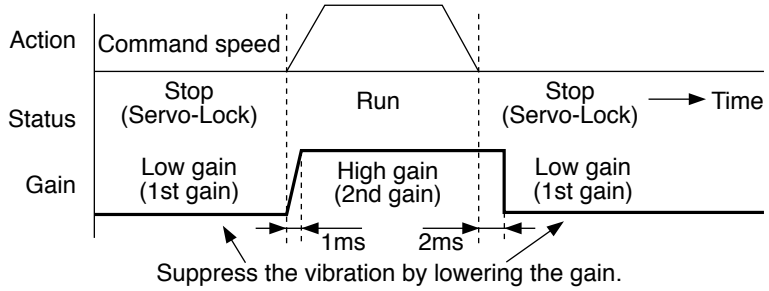
Related page • P.3-30 “Inputs and outputs on connector X4” • P.6-3 “Protective Function”

4. Manual Gain Tuning (Basic)

Gain Switching Function

By selecting appropriate gain based on internal data or external signal, the following effects can be obtained.

- Decrease the gain at the time of stoppage (servo lock) to reduce vibration.
- Increase the gain at the time of stoppage (setting) to shorten the settling time.
- Increase the gain during operation to improve command compliance.
- Based on condition of the equipment, change the gain with external signal.



<Example>

Following is the example when you want to reduce the noise at motor in stall (Servo-Lock), by setting up to lower gain after the motor stops.

- Make adjustment referring to the basic gain parameter setup table (P.5-9) as well.

Parameter No. (Pr□□)	Title of parameter	Execute manual gain-tuning without gain switching	Set up the same value as Pr1.05 to 1.09 (1st gain) to Pr1.00 to 1.04 (2nd gain)	Set up Pr1.05 to 1.09 (Gain switching condition)	Adjust Pr1.01 and 1.04 at stopping (1st gain)
1.00	1st gain of position loop	630			
1.01	1st gain of velocity loop	350			270
1.02	1st time constant of velocity integration	160			
1.03	1st filter of velocity detection	0			
1.04	1st time constant of torque filter	65			84
1.10	Velocity feed forward	300			
1.11	Filter of velocity feed forward	50			
1.05	2nd gain of position loop		630		
1.06	2nd gain of velocity loop		350		
1.07	2nd time constant of velocity integration		160		
1.08	2nd filter of velocity detection		0		
1.09	2nd time constant of torque filter time		65		
1.14	Action setup of 2nd gain	0		1	
1.15	1st mode of control switching			7	
1.16	1st delay time of control switching			30	
1.17	1st level of control switching			0	
1.18	1st hysteresis of control switching			0	
1.19	Switching time of position gain			0	
0.04	Inertia ration	• Enter the known value from load calculation • Measure the inertia ratio by executing nor mal auto-gain tuning • Default is 250			

Related page • P.4-5... “Details of parameter”

Setup of Gain Switching Condition

• Positing control mode, Full-closed control mode (◯: Corresponding parameter is valid, – : invalid)

Setup of gain switching condition			Setup parameters at position control, full-closed control		
			Delay time *1	Level	Hysteresis *2
Pr1.15	Switching condition to 2nd gain	Fig.	Pr1.16	Pr1.17	Pr1.18
0	Fixed to 1st gain		–	–	–
1	Fixed to 2nd gain		–	–	–
2	Gain switching input		–	–	–
3	Torque command	A	◯	◯ [%]	◯ [%]
4	Invalid (Fixed to 1st gain)		–	–	–
5	Speed command	C	◯	◯ [r/min]	◯ [r/min]
6	Position deviation	D	◯	◯ *3 [pulse]	◯ *3 [pulse]
7	Position command exists.	E	◯	–	–
8	Not in positioning complete	F	◯	–	–
9	Speed	C	◯	◯ [r/min]	◯ [r/min]
10	Command exists + velocity	G	◯	◯ [r/min] *5	◯ [r/min] *5

• Velocity control mode

Setup of gain switching condition			Setup parameters at velocity control mode		
			Delay time *1	Level	Hysteresis *2
Pr1.20	Switching condition to 2nd gain	Fig.	Pr1.16, 1.21	Pr1.17, 1.22	Pr1.18, 1.23
0	Fixed to 1st gain		–	–	–
1	Fixed to 2nd gain		–	–	–
2	Gain switching input		–	–	–
3	Torque command	A	◯	◯ [%]	◯ [%]
4	Variation of speed command is large.	B	–	◯ *4 [10(r/min)/s]	◯ *4 [10(r/min)/s]
5	Speed command	C	◯	◯ [r/min]	◯ [r/min]

• Torque control mode

Setup of gain switching condition			Setup parameters at torque control mode		
			Delay time *1	Level	Hysteresis *2
Pr1.24	Switching condition to 2nd gain	Fig.	Pr1.16, 1.25	Pr1.17, 1.26	Pr1.18, 1.27
0	Fixed to 1st gain		–	–	–
1	Fixed to 2nd gain		–	–	–
2	Gain switching input, GAIN ON		–	–	–
3	Variation of torque command is large.	A	◯	◯ [%]	◯ [%]

*1 Delay time (Pr1.16, 1.12 and 1.25) will be valid only when returning from 2nd to 1st gain.

*2 Hysteresis (Pr1.18, 1.23 and 1.27) is defined as the fig. below shows.

*3 Designate with either the encoder resolution or the external scale resolution depending on the control mode.

*4 When you make it a condition that there is speed variation of 10r/min in 1s, set up the value to 1.

*5 When Pr1.15=10, the meanings of delay time, level and hysteresis are different from the normal. (refer to Fig. G)

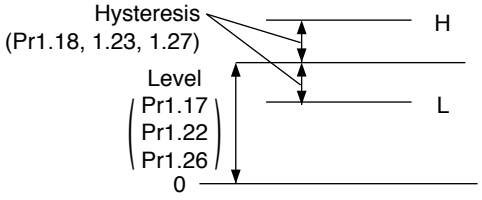


Fig.A

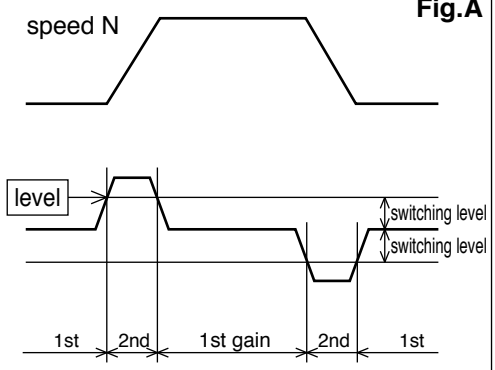


Fig. B

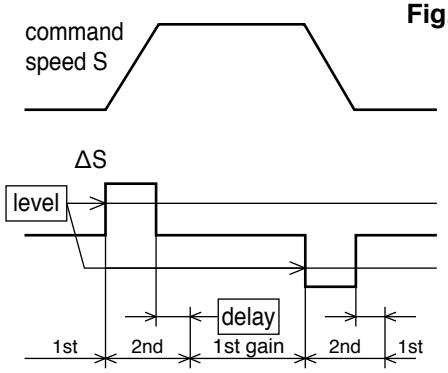


Fig. C

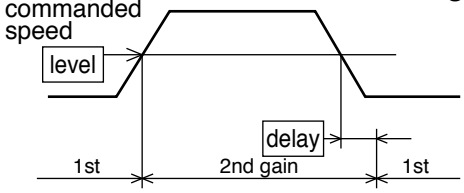


Fig. D

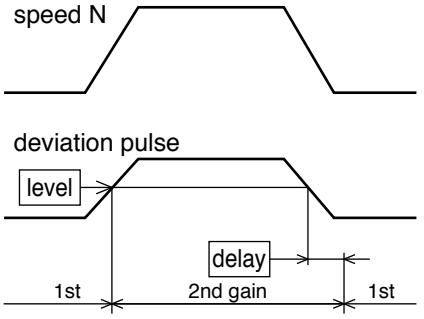


Fig. E

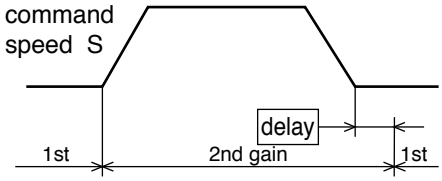


Fig. F

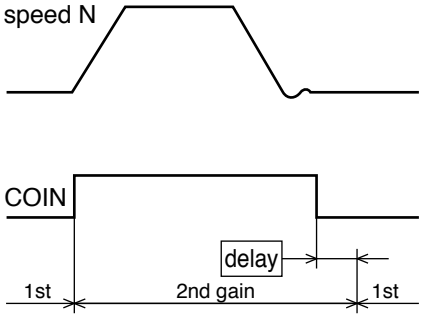
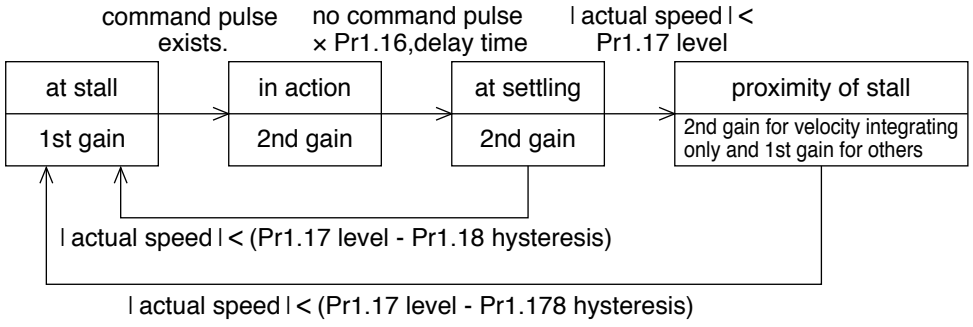


Fig. G



Caution Above Fig. does not reflect a timing lag of gain switching due to hysteresis (Pr1.18, 1.23 and 1.27).

4. Manual Gain Tuning (Basic)

Suppression of Machine Resonance

In case of a low machine stiffness, you cannot set up a higher gain because vibration and noise occur due to oscillation caused by axis distortion or other causes. By suppressing the resonance peak at the notch filter, higher gain can be obtained or the level of vibration can be lowered.

1. Torque command filter (Pr1.04 and Pr1.09)

Sets up the filter time constant so as to damp the frequency at vicinity of resonance frequency. You can obtain the cut off frequency of the torque command filter in the following formula.

Cut off frequency (Hz) fc = 1 / (2π x parameter setup value x 0.00001)

2. Notch filter (Pr2.00, 2.07 to Pr2.12)

• Adaptive filter

MINASA-5 series feature the adaptive filter. With this filter you can control vibration of the load which resonance points vary by machine by machine and normal notch filter or torque filter cannot respond. The adaptive filter is validated by setting up Pr23 (Adaptive filter mode setup) to 1. Enter the action command with Pr2.00 Adaptive filter mode set to a value other than 0.

If the resonance point affects the motor speed, parameters of 3rd notch filter and/or 4th notch filters are automatically set according to the number of adaptive filters.

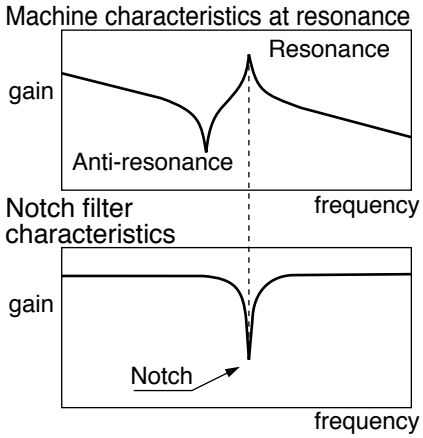
Pr2.00	Adaptive filter mode	1: Adaptive filter is valid
Pr2.07	3rd notch frequency	In no resonance point is found, the frequency is set to 5000.
Pr2.08	3rd notch width selection	Automatically set when the adaptive filter is active.
Pr2.09	3rd notch depth selection	
Pr2.10	4th notch frequency	Notch frequency is automatically set to the 2nd resonance frequency estimated by the adaptive filter. In no resonance point is found, the frequency is set to 5000.
Pr2.11	4th notch width selection	Automatically set when 2 adaptive filters are active.
Pr2.12	4th notch depth selection	

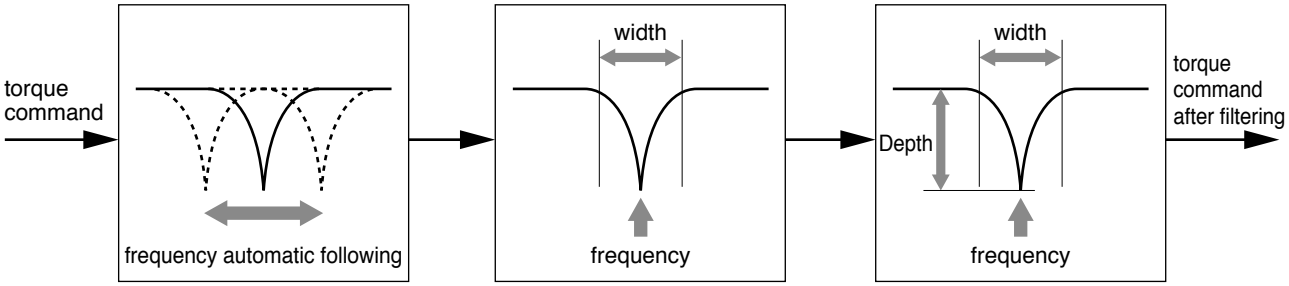
• Notch filter (Pr2.01 to 2.12)

MINASA-5 series feature 4 normal notch filters. You can adjust frequency and width and depth.

Pr2.01	1st notch frequency	Set the center frequency of the 1st notch filter. *1
Pr2.02	1st notch width selection	Set the width of notch at the center frequency of the 1st notch filter.
Pr2.03	1st notch depth selection	Set the depth of notch at the center frequency of the 1st notch filter.
Pr2.04	2nd notch frequency	Set the center frequency of the 2nd notch filter. *1
Pr2.05	2nd notch width selection	Set the width of notch at the center frequency of the 2nd notch filter.
Pr2.06	2nd notch depth selection	Set the depth of notch at the center frequency of the 2nd notch filter.
Pr2.07	3rd notch frequency	Set the center frequency of the 3rd notch filter. *1
Pr2.08	3rd notch width selection	Set the width of notch at the center frequency of the 3rd notch filter.
Pr2.09	3rd notch depth selection	Set the depth of notch at the center frequency of the 3rd notch filter.
Pr2.10	4th notch frequency	Set the center frequency of the 4th notch filter. *1
Pr2.11	4th notch width selection	Set the width of notch at the center frequency of the 4th notch filter.
Pr2.12	4th notch depth selection	Set the depth of notch at the center frequency of the 4th notch filter.

*1 The notch filter function will be invalidated by setting up this parameter to "5000".





Adaptive filter

Suppress resonance point instantaneously.

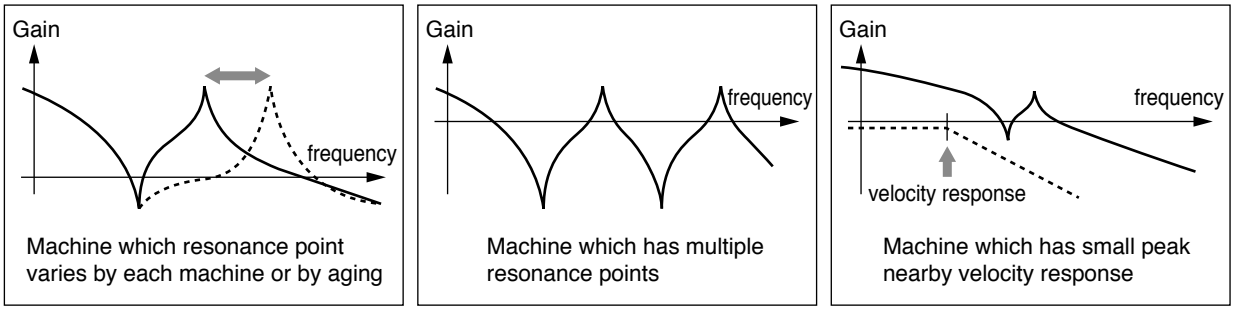
1st notch filter

Copying of the setup from the adaptive filter to 1st notch filter is enabled.

2nd to 4th notch filter

Adjustment of frequency, width and depth is enabled. (refer to P.5-22)

Example of application machine



daptive filter enabled

Adaptive filter and notch filter enabled

Depth adjustment enabled

Related page • P.4-12... , P.4-19... “Details of parameter”

Notch width and depth

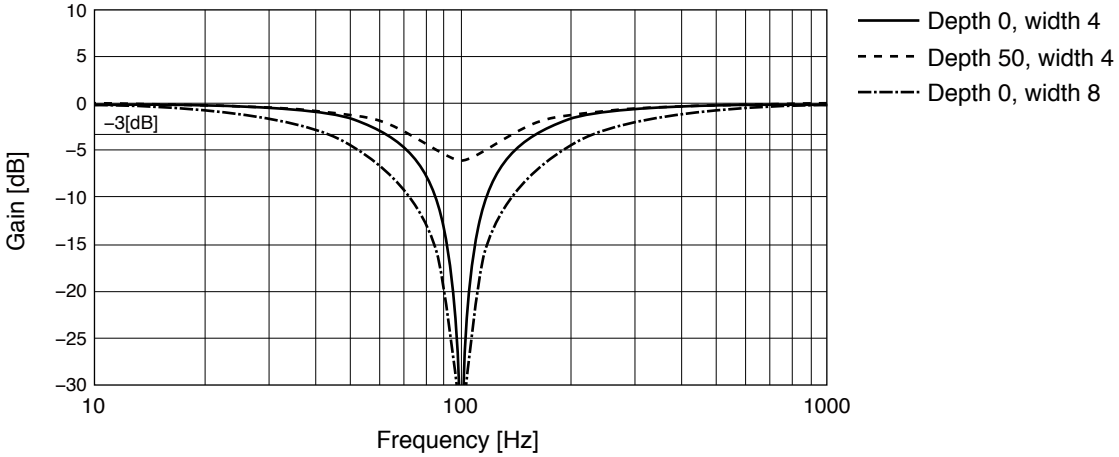
The width of the notch filter is the ratio of the width of –3dB attenuation frequency band with respect to the notch frequency at its center when depth is 0, and the value is as shown in the table below.

The notch filter depth indicates I:O ratio where the input at the center frequency is completely shut with setup value 0 but fully received with setup value 100. The table below shows this value in dB on the right.

Notch width	Band width/center frequency	
	A4 series (reference)	A5 series
0	0.41	0.5
1	0.56	0.59
2	0.71	0.71
3	0.86	0.84
4	1.01	1
5	—	1.19
6	—	1.41
7	—	1.68
8	—	2
9	—	2.38
10	—	2.83
11	—	3.36
12	—	4
13	—	4.76
14	—	5.66
15	—	6.73
16	—	8
17	—	9.51
18	—	11.31
19	—	13.45
20	—	16

Notch depth	I/O ratio	[dB]
0	0	–∞
1	0.01	–40
2	0.02	–34
3	0.03	–30.5
4	0.04	–28
5	0.05	–26
6	0.06	–24.4
7	0.07	–23.1
8	0.08	–21.9
9	0.09	–20.9
10	0.1	–20
15	0.15	–16.5
20	0.2	–14
25	0.25	–12
30	0.3	–10.5
35	0.35	–9.1
40	0.4	–8
45	0.45	–6.9
50	0.5	–6
60	0.6	–4.4
70	0.7	–3.1
80	0.8	–1.9
90	0.9	–0.9
100	1	0

Notch filter frequency characteristics



How to Check the Resonance Frequency of the Machine

(1) Start up the Setup Support Software, "PANATERM" and bring the frequency characteristics measurement screen.
(2) Set up the parameters and measurement conditions. (Following values are standard.)
- Set up Pr1.01 (1st gain of velocity loop) to 25 or so. (to lower the gain and make it easy to identify the resonance frequency)
- Set up the amplitude to 50 (r/min) or so. (not to saturate the torque)
- Make the offset to 100 (r/min) or so. (to increase the speed detecting data and to avoid the measurement error in the vicinity of speed-zero)
- Polarity is made positive direction with "+" and negative direction with "–".
- Setup the sampling rate to 0. (setup range to be 0 to 7.)

(3) Execute the frequency characteristic analysis.

Remarks
- Make sure that the revolution does not exceed the travel limit before the measurement.
  Standard revolutions are,
  Offset (r/min) × 0.017 × (sampling rate +1)
  Larger the offset, better measurement result you can obtain, however, revolutions may be increased.
- Set up Pr2.00 (Setup of adaptive filter mode) to 0 while you make measurement.

Note
- When you set a larger value of offset than the amplitude setup and make the motor run to the one direction at all time, you can obtain a better measurement result.
- Set up a smaller sampling rate when you measure a high frequency band, and a larger sampling rate when you measure a low frequency band in order to obtain a better measurement result.
- When you set a larger amplitude, you can obtain a better measurement result, but noise will be larger. Start a measurement from 50 [r/min] and gradually increase it.

Relation of Gain Adjustment and Machine Stiffness

In order to enhance the machine stiffness,
(1) Install the base of the machine firmly, and assemble them without looseness.
(2) Use a coupling designed exclusively for servo application with high stiffness.
(3) Use a wider timing belt. Belt tension to be within the permissible load to the motor shaft.
(4) Use a gear reducer with small backlash.
- Inherent vibration (resonance frequency) of the machine system has a large effect to the gain adjustment of the servo.
  You cannot setup a higher response of the servo system to the machine with a low resonance frequency (machine stiffness is low).

Note Please download the Setup support software “PANATERM”from our web site anduse after install to the PC. **http://industrial.panasonic.com/jp/i/fa_motor.html**

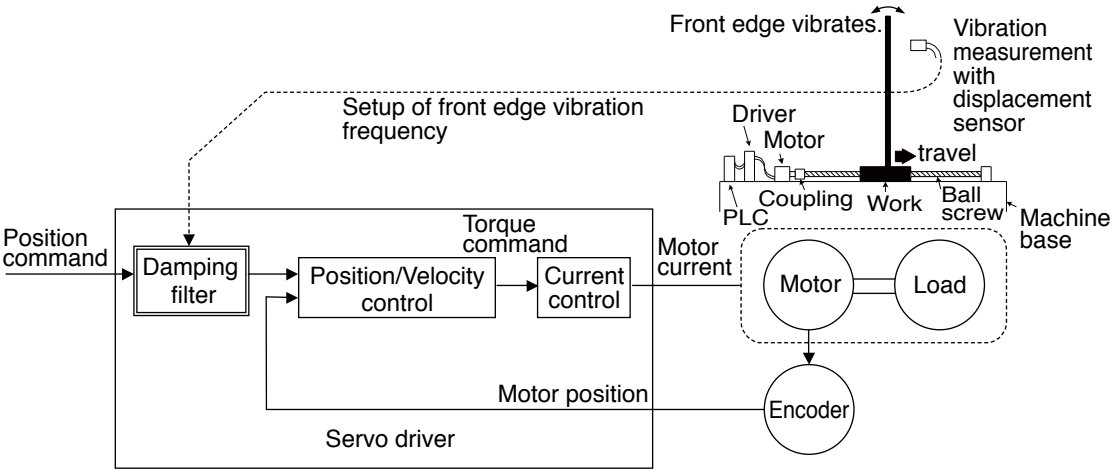
Related page • P.7-26 “Outline of Setup support software “PANATERM”

5. Manual Gain Tuning (Application)

Damping Control

Outline

This function reduces the vibration at the top or on whole of the equipment by removing the vibration frequency components specified by the positional command. Up to 2 among 4 frequency settings can be used at the same time.



Applicable Range

This function can only be applicable when the following conditions are satisfied.

	Conditions under which the damping control is activated
Control mode	• Control mode to be either or both position control or/and full-closed control. Pr02 = 0 : Position control Pr02 = 3 : 1st control mode of position and velocity control Pr02 = 4 : 1st control mode of position control and torque control Pr02 = 6 : Full-closed control

Caution

This function does not work properly or no effect is obtained under the following conditions.

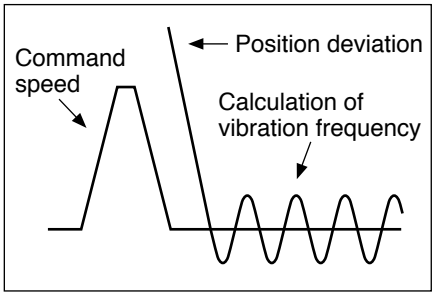
	Conditions which obstruct the damping control effect
Load	• Vibration is triggered by other factors than command (such as disturbance). • Ratio of resonance frequency and anti-resonance frequency is large. • Vibration frequency is out of the range of 10.0-200.0 [Hz].

Related page • P.4-4 “Details of parameter” • P.7-26 “Outline of Setup support software “PANATERM”

How to Use

(1) Setup of damping frequency (1st: Pr2.14, 2nd: Pr2.16, 3rd: Pr2.18, 4th: Pr2.20))

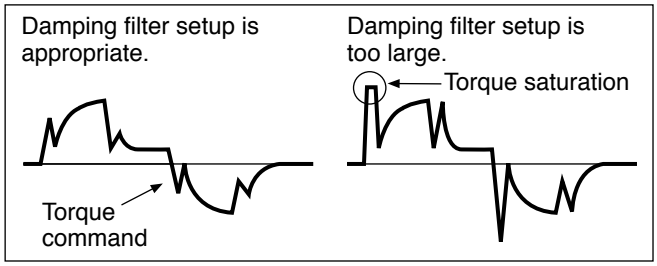
Measure the vibration frequency of the front edge of the machine. When you use such instrument as laser displacement meter, and can directly measure the load end vibration, read out the vibration frequency from the measured waveform and enter it.



(2) Setup of damping filter (1st: Pr2.15, 2nd: Pr2.17, 3rd: Pr2.19, 4th: Pr2.21))

First, set up 0.

You can reduce the settling time by setting up larger value, however, the torque ripple increases at the command changing point as the right fig. shows. Setup within the range where no torque saturation occurs under the actual condition. If torque saturation occurs, damping control effect will be lost.



(3) Setup of damping filter switching selection (Pr2.13)

You can switch the 1st or the 2nd damping filter depending on the vibration condition of the machine.

Pr2.13	VS-SEL2	VS-SEL1	1st damping	2nd damping	3rd damping	4th damping
0	—	—	○	○		
1	—	OFF	○		○	
	—	ON		○		○
2	OFF	OFF	○			
	OFF	ON		○		
	ON	OFF			○	
	ON	ON				○

Pr2.13	Position command direction	1st damping	2nd damping	3rd damping	4th damping
3	Positive direction	○		○	
	Negative direction		○		○

5. Manual Gain Tuning (Application)

Feed forward function

Outline

When position control or full closed control is used, positional deviation can be further reduced when compared with deviation where control is made only by feedback, and response is also improved, by calculating the velocity control command necessary for operation based on the internal positional command, and by adding velocity feed forward to the speed command calculated by comparison with position feedback.
The response time of the velocity control system is also improved by calculating torque command necessary for operation based on the velocity control command and by adding torque feed forward calculated by comparison with velocity feedback to the torque command.

Related Parameter

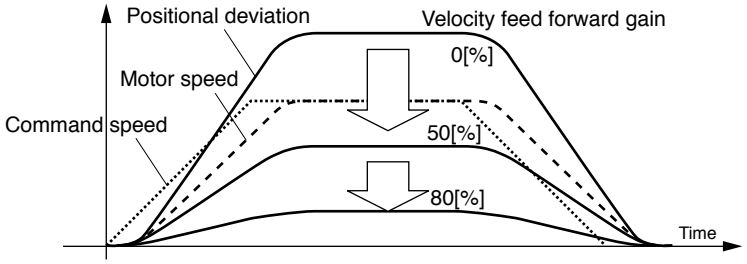
For A 5 series, the velocity feed forward and torque feed forward can be used.

Class	No.	Title	Function
1	10	Velocity feed forward gain	Multiply the velocity control command calculated according to the internal positional command by the ratio of this parameter and add the result to the speed command resulting from the positional control process.
1	11	Velocity feed forward filter	Set the time constant of 1st delay filter which affects the input of velocity feed forward.
1	12	Torque feed forward gain	Multiply the torque command calculated according to the velocity control command by the ratio of this parameter and add the result to the torque command resulting from the velocity control process.
1	13	Torque feed forward filter	Set up the time constant of 1st delay filter which affects the input of torque feed forward.
6	0	Analog torque feed forward conversion gain	Set the input gain of analog torque feed forward. 0 to 9 are invalid.
6	10	Function expansion setup	Set up the function in unit of bit. bit5 0: Analog torque feed forward is invalid. 1: Analog torque feed forward is Valid. * bit 0 = LSB

Usage example of velocity feed forward

The velocity feed forward will become effective as the velocity feed forward gain is gradually increased with the velocity feed forward filter set at approx. 50 (0.5 ms). The positional deviation during operation at a constant velocity is reduced as shown in the equation below in proportion to the value of velocity feed forward gain.

Positional deviation [unit of command] = command speed [unit of command/s] / positional loop gain [1/s] × (100 - velocity feed forward gain [%]) / 100

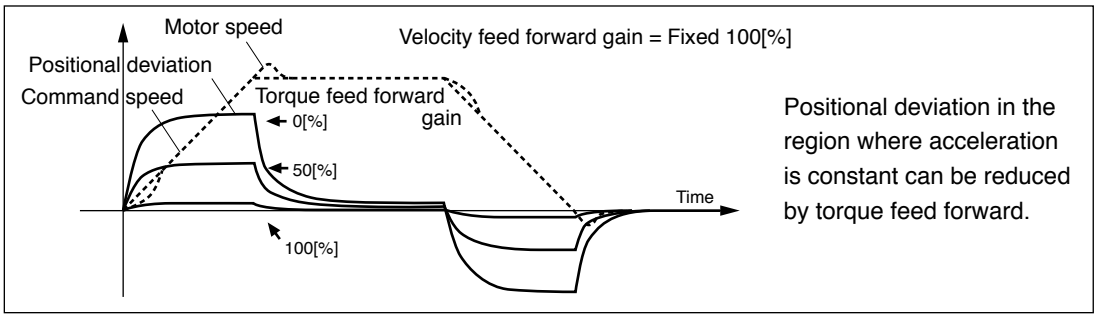


Positional deviation within the constant speed range will reduce as the velocity forward gain is increased.

With the gain set at 100%, calculatory positional deviation is 0, but significant overshoot occurs during acceleration/deceleration.
If the updating cycle of the positional command input is longer than the driver control cycle, or the pulse frequency varies, the operating noise may increase while the velocity feed forward is active. If this is the case, use positional command filter (1st delay or FIR smoothing), or increase the velocity forward filter setup value.

Usage example of torque feed forward

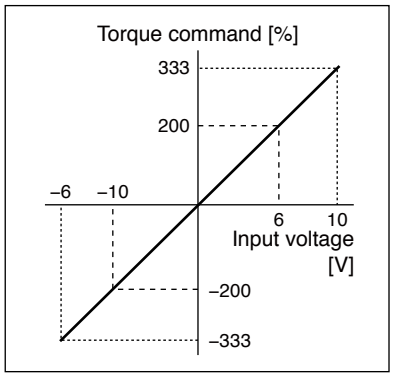
- To use the torque feed forward, correctly set the inertia ratio.
  Use the value that was determined at the start of the real time auto tuning, or set the inertia ratio that can be calculated from the machine specification to Pr0.04 Inertia ratio.
- The torque feed forward will become effective as the torque feed forward gain is gradually increased with the torque feed forward filter is set at approx. 50 (0.5 ms).
- Positional deviation at a constant acceleration/deceleration can be minimized close to 0 by increasing the torque forward gain. This means that positional deviation can be maintained at near 0 over entire operation range while driving in trapezoidal speed pattern under ideal condition where disturbance torque is not active .



Zero positional deviation is impossible in actual situation because of disturbance torque. As with the velocity feed forward, large torque feed forward filter time constant decreases the operating noise but increases positional deviation at acceleration change point.

Usage example of analog torque feed forward

- Setting bit 5 place of Pr6.10 Function expansion setup to 1 enables the analog torque feed forward. When the analog input 3 is used by another function (e.g. analog torque limit), the function becomes invalid.
- The voltage (V) applied to the analog input 3 is converted to the torque via Pr6.00 Analog torque feed forward conversion gain setup and added to the torque command (%).
- The voltage (V) applied to the analog input 3 is converted to the motor torque command (%) through the process as shown in the graph below.
- The slope represents when Pr6.00 = 30. The slope changes as the setup value changes.



Torque command (%) = 100 × input voltage (V) / (Pr6.00 setup value × 0.1)

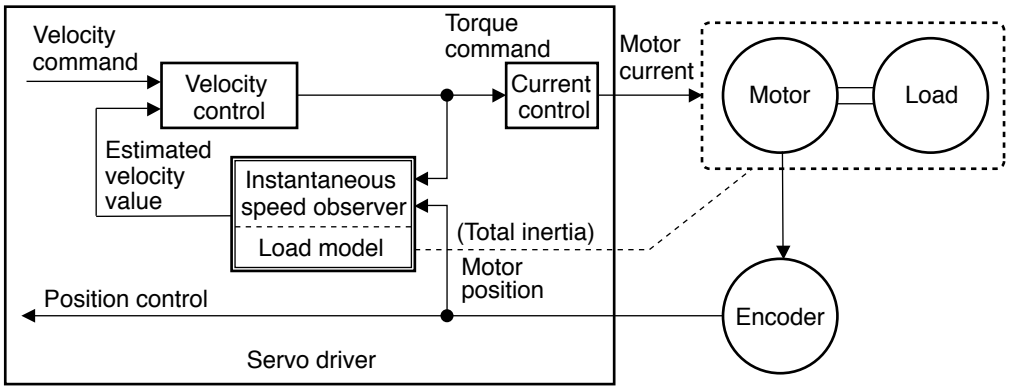
Related page • P.4-13... “Details of parameter”

5. Manual Gain Tuning (Application)

Instantaneous Speed Observer

Outline

This function enables both realization of high response and reduction of vibration at stopping, by estimating the motor speed using a load model, hence improving the accuracy of the speed detection.



Applicable Range

This function can be applicable only when the following conditions are satisfied.

	Conditions under which the instantaneous speed observer is activated
Control mode	• Control mode to be either or both position control or/and velocity control. Pr0.01 = 0 : Position control Pr0.01 = 1 : Velocity control
Others	• Should be in servo-on condition • Input signals such as the deviation counter clear and command input inhibit, and parameters except for controls such as torque limit setup, are correctly set, assuring that the motor can run smoothly. • Real-time auto-tuning should be disabled. (Pr0.02=0)

Caution

This function does not work properly or no effect is obtained under the following conditions.

	Conditions which obstruct instantaneous speed observer action
Load	• Gap between the estimated total load inertia (motor + load) and actual machine is large. e.g.) Large resonance point exists in frequency band of 300[Hz] or below. Non-linear factor such as large backlash exists. • Load inertia varies. • Disturbance torque with harmonic component is applied.
Others	• Settling range is very small.

Related page • P.4-52 “Details of parameter”

Related Parameter

Class	No.	Title	Function
6	10	Function expansion setup	Speed observer enable bit (bit 0) valid/invalid the function. bit0 0: Invalid 1: Valid * bit 0 = LSB

How to Use

(1) Setup of inertia ratio (Pr0.04)

Set up as exact inertia ratio as possible.

- When the inertia ratio (Pr20) is already obtained through real-time auto-gain tuning and is applicable at normal position control, use this value as Pr20 setup value.
- When the inertia ratio is already known through calculation, enter this calculated value.
- When the inertia ration is not known, execute the normal mode auto-gain tuning and measure the inertia ratio.

(2) Adjustment at normal position control

- Adjust the position loop gain, velocity loop gain, etc.

(3) Setup of instantaneous velocity observer (Pr6.10)

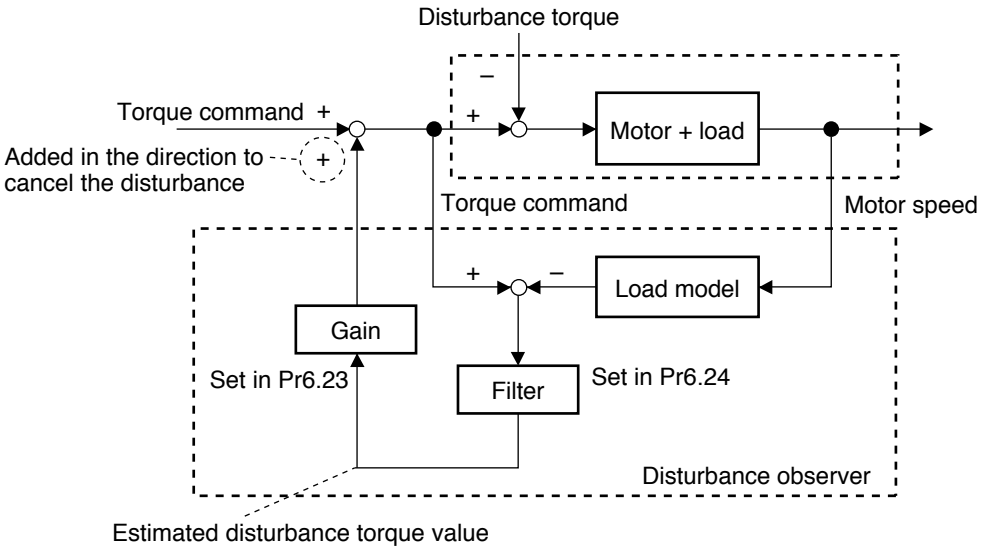
- By enabling instantaneous speed observer function through function expansion setup (Pr6.10), the speed detection method changes to the instantaneous speed observer.
- When you experience a large variation of the torque waveform or noise, return this to 0, and reconfirm the above cautions and (1).
- When you obtain the effect such as a reduction of the variation of the torque waveform and noise, search an optimum setup by making a fine adjustment of Pr0.04 (Inertia ratio) while observing the position deviation waveform and actual speed waveform to obtained the least variation. If you change the position loop gain and velocity loop gain, the optimum value of the inertia ratio (Pr0.04) might have been changed, and you need to make a fine adjustment again.

5. Manual Gain Tuning (Application)

Disturbance observer

Outline

This function uses the disturbance torque determined by the disturbance observer to reduce effect of disturbance torque and vibration.



Applicable Range

This function can be applicable only when the following conditions are satisfied.

	Conditions under which the disturbance observer is activated
Control mode	• Control mode to be either or both position control or/and velocity control. Pr0.01 = 0 : Position control Pr0.01 = 1 : Velocity control
Others	• Should be in servo-on condition • Input signals such as the deviation counter clear and command input inhibit, and parameters except for controls such as torque limit setup, are correctly set, assuring that the motor can run smoothly. • Real-time auto-tuning should be disabled. (Pr0.02=0) • Instantaneous speed observer should be disabled. (Pr6.10=0) • Gain switchover should be enabled.

Caution

Effect may not be expected in the following condition.

	Conditions which obstruct disturbance observer action
Load	• Resonant frequency is lower than the cutoff frequency estimated by the disturbance observer. • Disturbance torque contains many high frequency components.

Related page • P.4-4... , P.4-52 “Details of parameter”

Related Parameter

Class	No.	Title	Function
6	10	Function expansion setup	Sets bits related to disturbance observer. bit1 0: Invalid 1: Valid bit2 0: Always valid 1: alid only when 1st gain is selected. * bit 0 = LSB Example) To use the disturbance observer in the enabled mode only when 1st gain is selected: Setup value = 6 To use the disturbance observer always in the enabled mode: Setup value = 2
6	23	Disturbance torque compensating gain	Set up compensating gain against disturbance torque.
6	24	Disturbance observer filter	Set up the filter time constant according to the disturbance torque compensation.

How to Use

1) With Pr6.10 Function enhancement setup, set observer enable/disable and operation mode (always enable/enable only when 1st gain is selected).

2) Setup of Pr6.24 (Disturbance observer filter)

First, set up Pr6.24 to a larger value and check the operation with Pr6.23 Disturbance torque compensating gain set to a low value, and then gradually decrease the setup value of Pr6.24. A low filter setup value assures disturbance torque estimation with small delay and effectively suppresses effects of disturbance. However, this results in larger operation noise. Well balanced setup is required.

3) Setup of Pr6.23 (Disturbance torque compensating gain)

After setting up Pr6.24, increase Pr6.23.

The disturbance suppressing capability increases by increasing the gain, but it is associated with increasing volume of operation noise.

This means that well balanced setup can be obtained by adjusting Pr6.24 and Pr6.23.

5. Manual Gain Tuning (Application)

3rd gain switching function

Outline

In addition to the normal gain switching function described on P.5-17, 3rd gain switching function can be set to increase the gain just before stopping. The higher gain shortens positioning adjusting time.

Applicable Range

This function can be applicable only when the following conditions are satisfied.

	Conditions under which the 3rd gain switching function is activated
Control mode	• Control mode to be either or both position control or/and full-closed control. Pr0.01 = 0 : Position control Pr0.01 = 6 : Full-closed control
Others	• Should be in servo-on condition • Input signals such as the deviation counter clear and command input inhibit, and parameters except for controls such as torque limit setup, are correctly set, assuring that the motor can run smoothly.

Related Parameter

Class	No.	Title	Function
6	5	Position 3rd gain valid time	Set up the time at which 3rd gain becomes valid.
6	6	Position 3rd gain scale factor	Set up the 3rd gain by a multiplying factor of the 1st gain: 3rd gain = 1st gain × Pr6.06/100

Related page • P.4-4... , P.4-51 “Details of parameter”

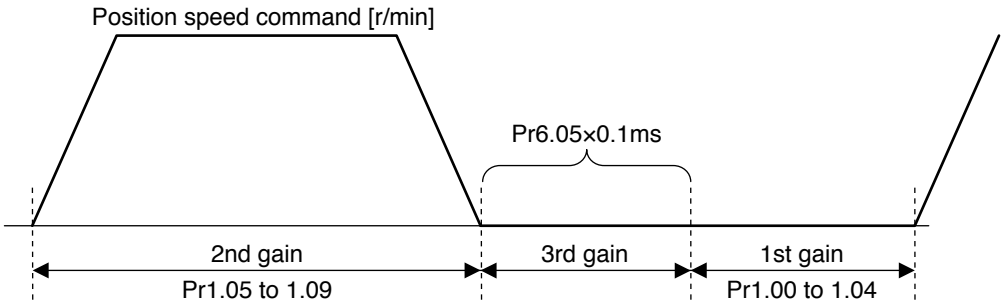
How to Use

While in the condition under which the normal gain switching functions, set the 3rd gain application time to Pr6.05 Position 3rd gain enable time, and set the 3rd gain (scale factor with reference to 1st gain) to Pr6.06 Position 3rd gain magnification ratio.

- If 3rd gain is not used, set Pr6.05 to 0 and Pr6.06 to 100.
- The 3rd gain is enabled only for position control or full closed control.
- During the 3rd gain period, only position loop gain/speed proportional gain becomes 3rd gain, during other periods, 1st gain setting is used.
- When the 2nd gain switching condition is established during 3rd gain period, 2nd gain is used.
- During transition from 2nd gain to 3rd gain, Pr1.19 Position gain switching time is applied.

Caution When the gain is switched from 2nd to 1st by the change in parameter, the 3rd gain period appears.

Example)
Pr1.15 Position control switching mode = 7 switching condition: with positional command:



[3rd gain period]
Position loop gain = Pr1.00 × Pr6.06/100
Speed proportional gain = Pr1.01 × Pr6.06/100
Time constant of velocity integration, speed detection filter and torque filter directly use the

5. Manual Gain Tuning (Application)

Friction torque compensation

Outline

To reduce effect of friction represented by mechanical system, 2 types of friction torque compensation can be applied: offset load compensation that cancels constant offset torque and the dynamic friction compensation that varies direction as the operating direction varies.

Applicable Range

This function can be applicable only when the following conditions are satisfied.

	Conditions under which the Friction torque compensation is activated
Control mode	• Specific to individual functions. Refer to "Related parameters" shown below.
Others	• Should be in servo-on condition • Input signals such as the deviation counter clear and command input inhibit, and parameters except for controls such as torque limit setup, are correctly set, assuring that the motor can run smoothly.

Related Parameter

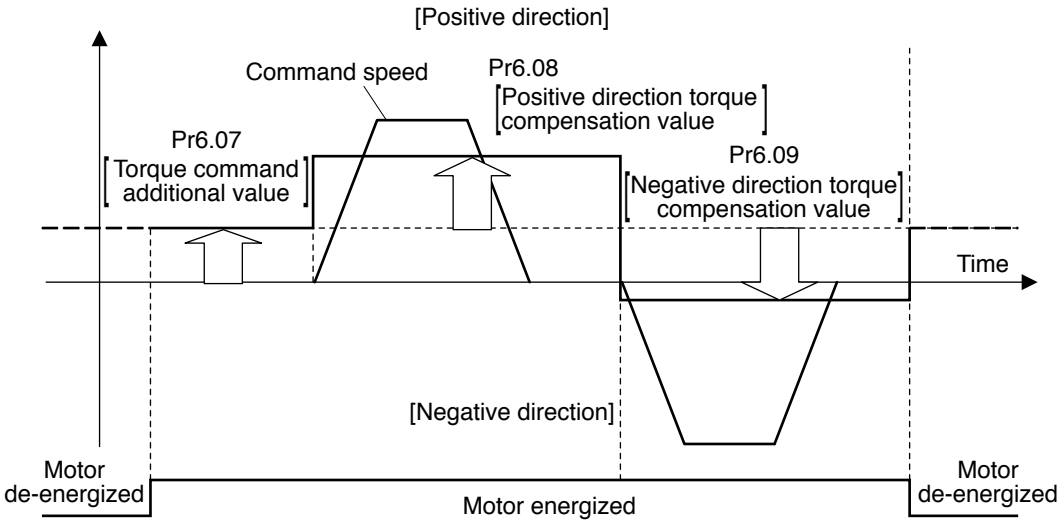
Combine the following 3 parameters to setup appropriate friction torque compensation.

Class	No.	Title	Function
6	7	Torque command additional value	Set up the offset load compensation value usually added to the torque command in a control mode except for the torque control mode.
6	8	Positive direction torque compensation value	Set up the dynamic friction compensation value to be added to the torque command when forward positional command is fed.
6	9	Negative direction torque compensation value	Set up the dynamic friction compensation value to be added to the torque command when negative direction positional command is fed.

Related page • P.4-51, P.4-52 “Details of parameter”

How to Use

The friction torque compensation will be added in response to the entered positional command direction as shown below.



The friction compensation torque is the sum of the offset load compensation value which is set according to the torque command additional value (always constant) and the dynamic friction compensation torque which is set according to positive/negative direction torque compensation value.
The command speed direction is reset upon power-up or when the motor is de-energized.

- Pr6.07 [Torque command additional value] reduces variations in positioning operation (performance is affected by direction of movement). These variations occur when constant offset torque resulting from weight on vertical axis is applied to the motor.
- Certain loads such as belt driven shaft requires high dynamic friction torque, which lengthens positioning setting time or varies positioning accuracy. These problems can be minimized by setting the friction torque of every rotating direction into individual parameters. Pr6.08 [Positive direction torque compensation value] and Pr6.09 [Negative direction torque compensation value] can be used for this purpose.

Caution The offset load compensation and dynamic friction compensation can be used individually or in combination. However, some control modes impose limit on application.

- For torque control: Offset load compensation and dynamic friction compensation are set at 0 regardless of parameter setting.
- For velocity control with servo-off: Offset load compensation per Pr6.07 is enabled. Dynamic friction compensation is set at 0 regardless of parameter setting.
- For position control or full closed control with servo-on: Previous offset load compensation and dynamic friction compensation values are maintained until the first positional command is applied where the offset load compensation value is updated according to Pr6.07. The dynamic friction compensation value is updated to parameters Pr.6.08 and Pr6.09 depending on command direction.

5. Manual Gain Tuning (Application)

Inertia ratio switching function

Outline

Inertia ratio can be switched between No.1 and No.2 by the switching input (J-SEL). This feature is useful in application where the load inertia changes in two steps.

Applicable Range

This function can be applicable only when the following conditions are satisfied.

	Conditions under which the Inertia ratio switching function is activated
Control mode	• Can be used in all control modes. Pr0.01 = 0 : Position control Pr0.01 = 1 : Velocity control Pr0.01 = 2 : Torque control Pr0.01 = 3 : Position/Velocity control Pr0.01 = 4 : Position/Torque control Pr0.01 = 5 : Velocity/Torque control Pr0.01 = 6 : Full-closed control
Others	• Should be in servo-on condition • Input signals such as the deviation counter clear and command input inhibit, and parameters except for controls such as torque limit setup, are correctly set, assuring that the motor can run smoothly. • Real-time auto-tuning should be disabled. (Pr0.02=0) • Adaptive filter should be disabled. (Pr2.00=0) • Instantaneous speed observer should be disabled. (Pr6.10=0) • Disturbance observer should be disabled. (Pr6.24=0 bit1=0)

Caution

- Be sure to change the inertia ratio while the motor is in stop state. Otherwise, vibration or oscillation will occur.
- If the difference between the 1st inertial ratio and 2nd inertia ratio is large, vibration, etc., may occur even in stop mode. These potential problems should be identified on the actual model.

Related page • P.3-30 “Inputs and outputs on connector X4” • P.4-4... , P.4-52, 53 “Details of parameter”

Related Parameter

Combine the following 3 parameters to setup appropriate inertia ratio switching function.

Class	No.	Title	Function
6	10	Function expansion setup	Sets bits related to inertia ratio switching function. bit1 0: Invalid 1: Valid bit2 0: Always valid 1: alid only when 1st gain is selected. * bit 0 = LSB Example) To enable inertial ratio switching Setup value = 8
0	04	Inertia ratio	Set 1st inertia ratio. You can set up the ratio of the load inertia against the rotor (of the motor) inertia.
6	13	2nd Inertia ratio	Set 1st inertia ratio. You can set up the ratio of the load inertia against the rotor (of the motor) inertia.

How to Use

- Selects 1st inertia ratio or 2nd inertia ratio according to the inertia ratio select input (J-SEL).

Inertia ratio switching input (J-SEL)	Applicable inertia ratio
OFF	1st Inertia ratio (Pr0.04)
ON	2nd Inertia ratio (Pr6.12)

5. Manual Gain Tuning (Application)

Hybrid vibration damping function

Outline

This function suppresses vibration due to amount of twist between the motor and load in the full closed control mode. This function enables high gain setting.

Outline

This function can be applicable only when the following conditions are satisfied.

	Conditions under which the Hybrid vibration damping function is activated
Control mode	• Full-closed control mode
Others	• Should be in servo-on condition • Input signals such as the deviation counter clear and command input inhibit, and parameters except for controls such as torque limit setup, are correctly set, assuring that the motor can run smoothly.

Caution

The effect of this function will be proportional to the amount of twist between the motor and load.

Related Parameter

Class	No.	Title	Function
6	34	Hybrid vibration suppression gain	Set up the hybrid vibration suppression gain for full-closed controlling. First set it to the value identical to that of poison loop gain, and then fine tune as necessary.
6	35	Hybrid vibration suppression filter	Set up the time constant of the hybrid vibration suppression filter for full-closed controlling.

How to Use

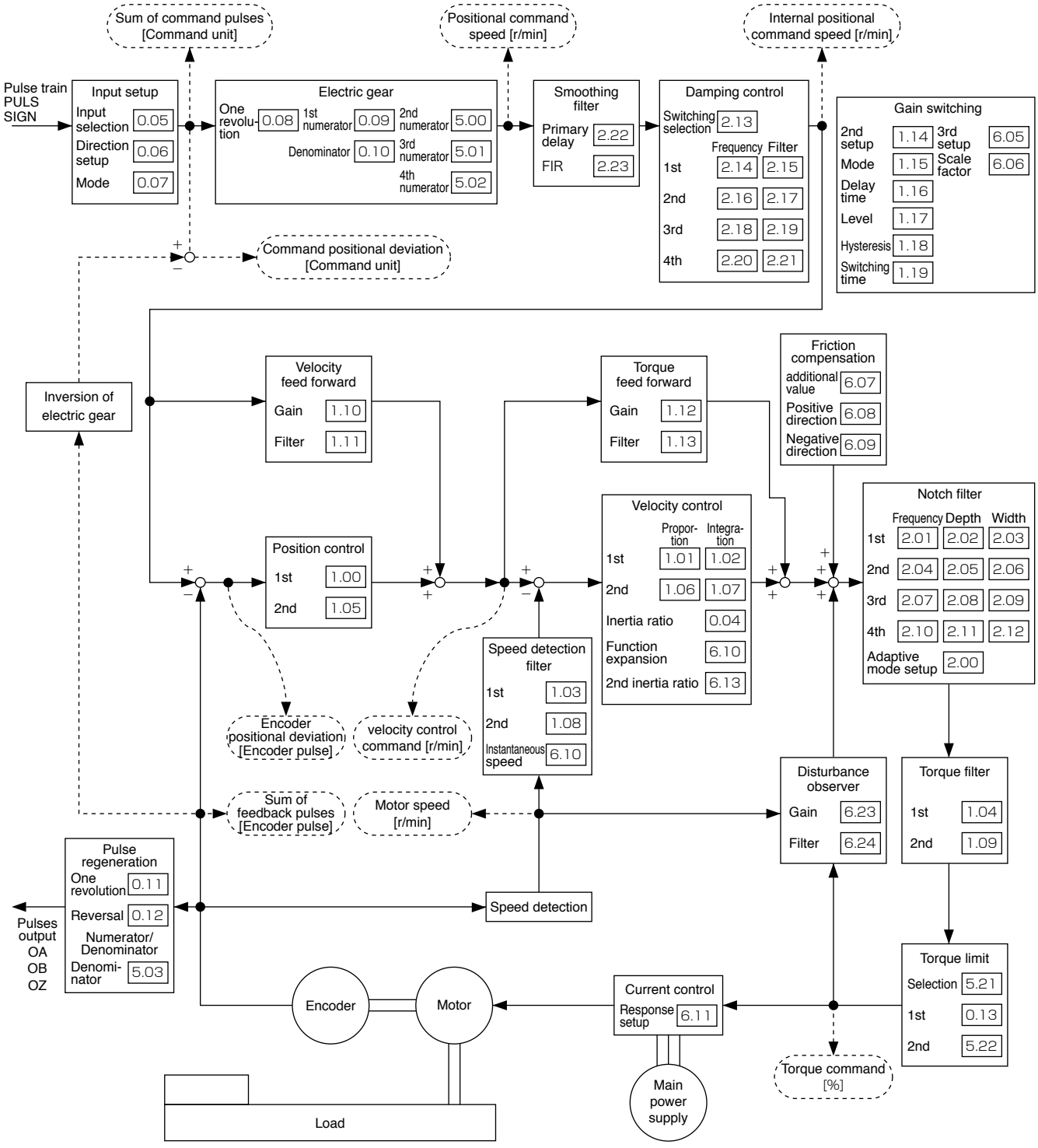
[1] Set Pr6.34 Hybrid vibration suppression gain to the value equal to that of positional loop gain.

[2] Driving under full closed control, gradually increase the setup value of Pr6.35 Hybrid vibration suppression filter while checking response change.
When the response is improved, adjust Pr6.34 and Pr6.35 to determine the combination that provides the best response.

Related page • P.4-57 “Details of parameter”

2. Control Block Diagram

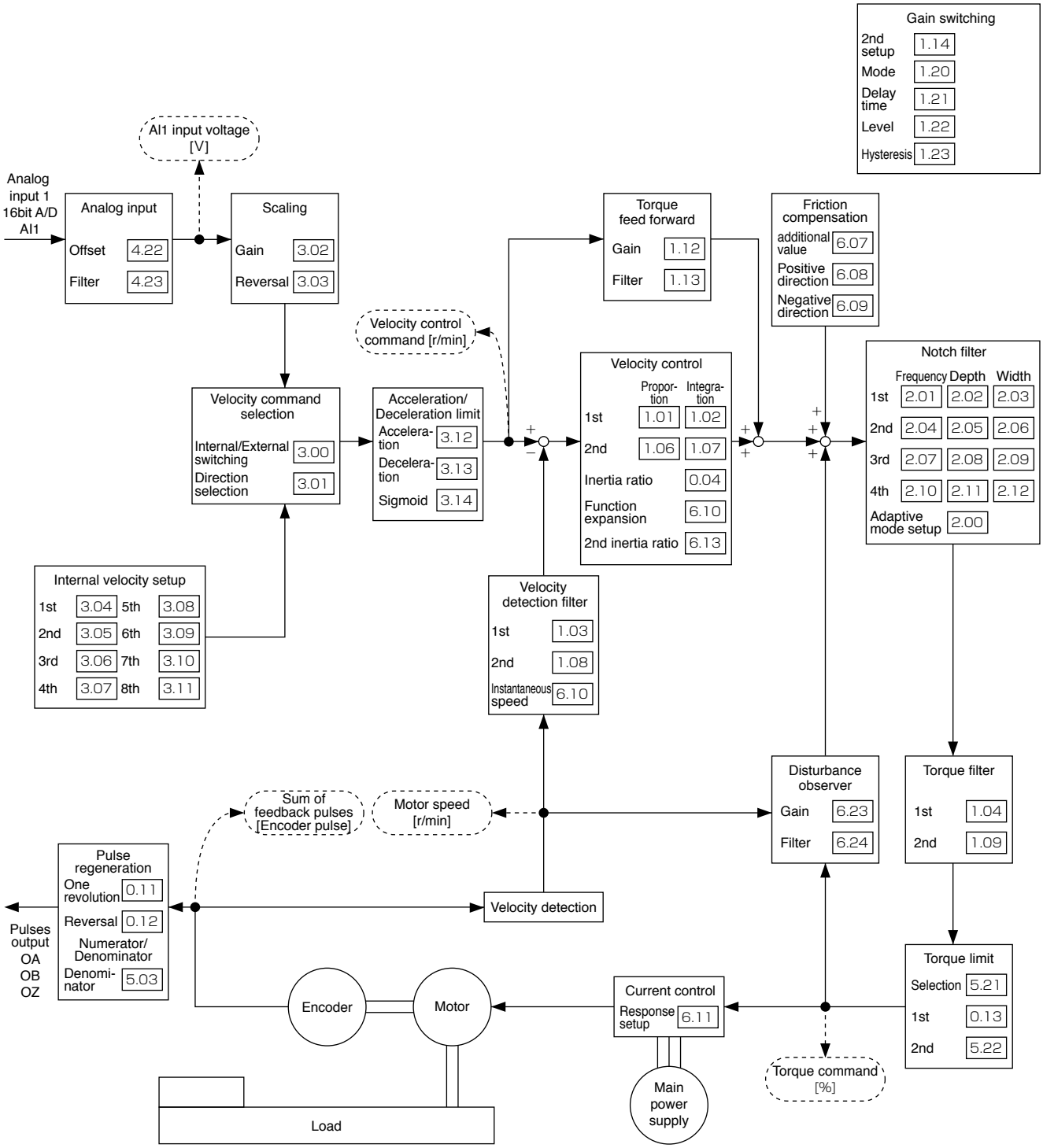
Position Control Mode



Related page
• P.3-18 “Wiring example of position control mode”
• P.3-20 “Connecting example to host controller”
• P.3-30 “Inputs and outputs on connector X4” • P.4-2 “List of Parameters”

2. Control Block Diagram

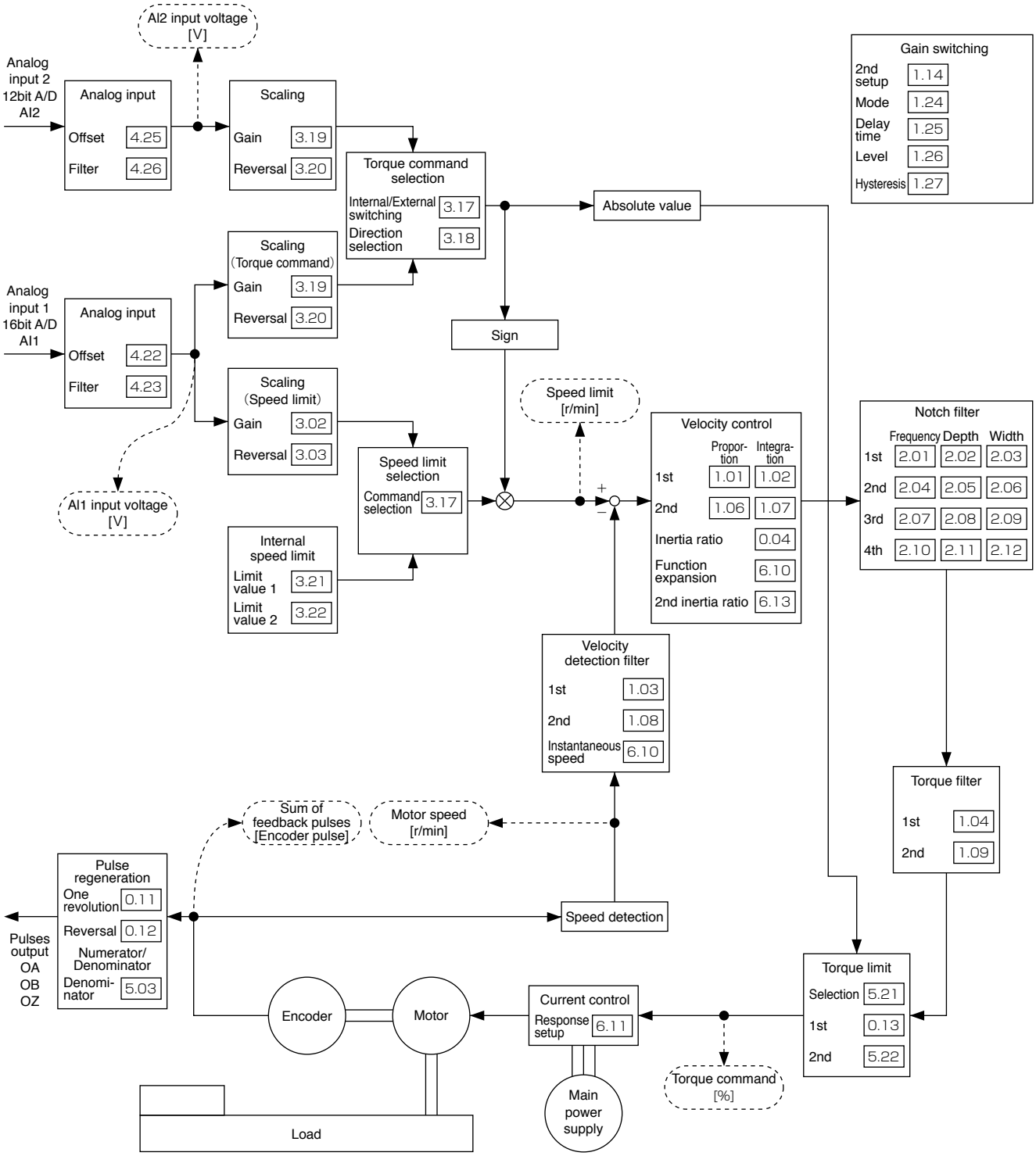
Velocity Control Mode



Related page • P.3-18 “Wiring example of velocity control mode”

2. Control Block Diagram

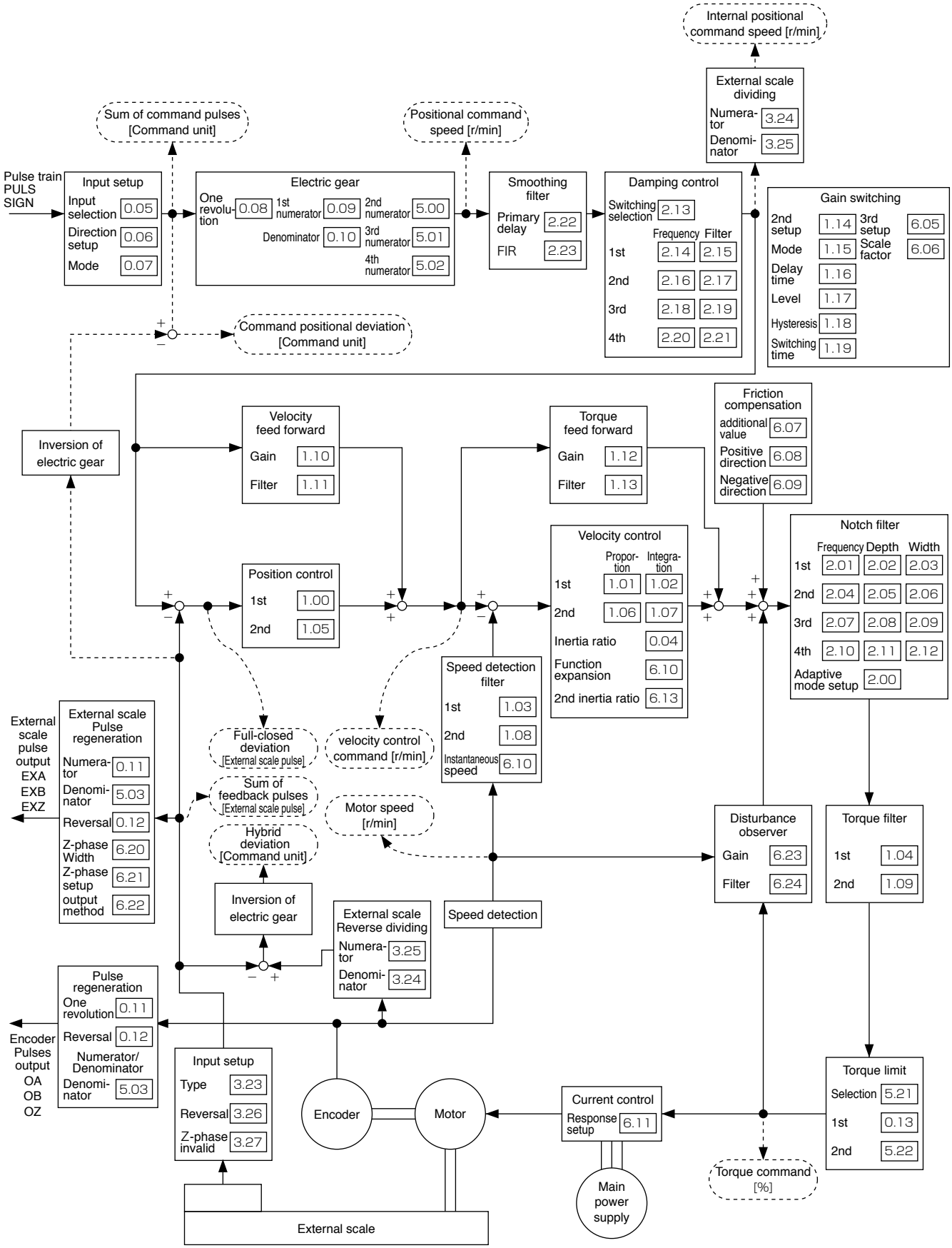
Torque Control Mode



Related page • P.3-19 “Wiring example of torque control mode”

2. Control Block Diagram

Full-closed Control Mode



Related page • P.3-19 “Wiring example of full-close control mode”